Night Signals

A public synthetic-data audit and internal model evaluation

Source and scope

The primary file contains 1,000 Kaggle records described by the uploader as synthetic. This project did not design or conduct a survey. No sampling frame, item wording, generation algorithm or within-night timestamps are provided. The results below describe this file; they are not estimates for a real population. Source: kaggle.com/datasets/harpartapsingh13/sleep-and-doomscrolling-habits-dataset. Local source checksum and field caveats are documented in data/PROVENANCE.md and DATA_DICTIONARY.md.

Target validity

A direct category-by-score check finds that 213 of 330 records labeled Poor have sleep_quality_score 5/5. The category derivation is undocumented. The classifier should be read as recovery of a synthetic file label, not detection of clinical poor sleep.

Research questions and exploratory tests

The five habit/category checks were specified after prior inspection and are exploratory, not preregistered. Pearson chi-square tests use complete cases; Benjamini-Hochberg FDR adjusts these five comparisons. Cramer's V is descriptive; statistical significance does not establish a protective effect.

Variable	n	Cramer's V	FDR p
bedtime routine type	980	0.097	0.0464
uses night mode	1000	0.034	0.705
keeps phone in bedroom	1000	0.060	0.273
consumes negative news content	1000	0.024	0.742
doomscroller	1000	0.457	2.35e-45

Routines and negative news

The unadjusted routine association is small. Reading does not stand out as protective: its Good share is below the 34% file-wide base rate. The no-fixed-routine group has the highest Good share. Negative-news exposure has a weak association with the category (chi-square p about 0.742). None of these contrasts establishes causality.

Routine	n	Good %	Poor %
Meditation/Journaling	108	36.1	38.0
No Fixed Routine	156	42.9	24.4
Reading	157	31.8	33.1
Scrolling Social Media	375	28.0	38.1
Watching Videos	184	37.0	29.3

Model protocol

The binary target is Poor versus all other categories. Measured sleep outcomes, quality score and record ID are excluded. Numeric imputation/scaling and categorical encoding occur in model pipelines. Five-by-three nested cross-validation compares candidates on 750 development records. A sigmoid-calibrated logistic ensemble and a cost-weighted threshold were fitted using development data. Earlier experiments may have inspected overlapping holdout rows, so prospective test independence is unverified. The cross-sectional source also cannot establish predictor availability at bedtime.

Internal evaluation

Selected logistic model nested-CV balanced accuracy: 0.737 (fold SD 0.041). Historical 250-row holdout: balanced accuracy 0.760 (bootstrap 95% interval 0.704-0.816), ROC AUC 0.825, PR AUC 0.720, Brier 0.155, sensitivity 0.723, specificity 0.796; threshold 0.335. These are internal label-recovery metrics.

Retrospective holdout calibration: intercept 0.149 (95% CI -0.231 to 0.529); slope 1.218 (95% CI 0.889 to 1.546). This is not external calibration.

Development-only comparator	Features	BA at 0.5	OOF ROC AUC
majority class	0	0.500	0.500
doomscroller label rule	1	0.713	0.713
bedtime minutes only	1	0.651	0.755
full minus self-label and phone checks	20	0.676	0.746
full minus self-label	21	0.682	0.757
full candidate features	22	0.694	0.775

The fixed-threshold logistic comparisons are diagnostic and are not directly equivalent to the tuned, calibrated nested-CV production estimate.

Independent context and limitations

NHANES includes 8,040 analytic adults. Age/sex-adjusted weighted WLS with stratum fixed effects and PSU-clustered covariance is an approximation, not full survey-design inference. PAD800/PAD820 are minutes per occasion; the observed activity comparison does not support a protective claim. PhysioNet subjects are unrelated to the Kaggle file. Neither dataset validates the classifier or its doomscrolling association. The high-exposure favorable-sleep subgroup is small, and contrasts in sleep duration, debt or fatigue are circular because those quantities track the outcome. External real-world model validation has not been performed.

Reproducibility: python scripts/audit_research_claims.py; python scripts/compare_simple_baselines.py; python scripts/train_production_model.py. The report is generated by scripts/build_current_report.py. Legacy figures remain historical and should not be cited as current-model results.